IUBAT – International University of Business Agriculture and Technology

Department of Computer Science and Engineering

AI-Based Expert System for Loan Approval Decision Support

Course: CSE 4383 – Artificial Intelligence and Expert Systems

Spring Semester 2026

Submitted By:

Abir Hasan
ID: 22303364 (Sl 04)
Section: M

Submitted To:

Dipta Mohon Das
Lecturer
Department of CSE, IUBAT

Submission Date: 23 May, 2026

Contents

1 Introduction

Artificial Intelligence (AI) has significantly transformed modern financial systems, particularly in areas such as credit evaluation, fraud detection, and loan approval. Traditional loan evaluation methods rely heavily on human judgment, which is often slow, inconsistent, and susceptible to implicit bias. Financial institutions are increasingly turning toward AI and Expert Systems (ES) to automate and improve the reliability of their decision-making processes.

Loan approval is a complex decision-making task that requires the evaluation of multiple applicant attributes, including monthly income, employment status, credit history, loan amount requested, loan term, marital status, and dependents. These factors often interact in non-linear and uncertain ways, making rule-based systems alone insufficient for capturing the full complexity of the problem.

This project presents a **hybrid AI-based expert system** that integrates three core AI techniques:

- **Rule-Based Reasoning** – encodes domain knowledge through explicit IF-THEN rules
- **Probabilistic Reasoning (Naive Bayes)** – handles uncertainty by computing posterior probabilities of loan approval
- **Machine Learning (Logistic Regression)** – learns patterns from historical data to improve prediction accuracy

The system is validated on a real-world publicly available dataset from Kaggle and evaluated using standard classification metrics. Additionally, the project addresses the ethical dimensions of automated financial decision-making, including bias, fairness, and transparency.

1.1 Motivation

Traditional loan approval processes suffer from several limitations:

- **Time-consuming**: Manual evaluation of each applicant can take days or weeks.
- **Human bias**: Loan officers may unconsciously favor or disfavor applicants based on non-financial factors.
- **Inconsistency**: Different officers may make different decisions for identical applicant profiles.
- **Scalability**: Manual processes cannot scale efficiently with increasing application volumes.

An AI-based expert system addresses these issues by providing consistent, explainable, and data-driven loan decisions. It can process thousands of applications rapidly while providing transparent reasoning for each outcome.

1.2 Objectives

The specific objectives of this project are:

1. Design and implement a hybrid AI-based expert system for loan approval prediction.
2. Apply probabilistic reasoning (Naive Bayes) to model uncertainty in decision-making.
3. Implement a rule-based knowledge base encoding domain expertise.
4. Integrate a machine learning model (Logistic Regression) to improve prediction accuracy.
5. Evaluate the system using accuracy, precision, recall, and F1-score.
6. Critically examine ethical concerns related to AI in financial decision-making.

1.3 Scope

The system is designed for binary loan approval decisions (Approved/Rejected). It takes structured tabular input data and produces a prediction along with an explanation of the key factors influencing the decision. The scope is limited to individual personal loans based on the features present in the chosen dataset.

2 Literature Review

2.1 AI in Financial Decision-Making

The application of AI to financial systems has been the subject of extensive research over the past two decades. Khandani et al. [1] demonstrated that machine learning models significantly outperform traditional credit scoring techniques such as FICO scores in predicting consumer loan defaults. Their work established a foundation for the use of ensemble methods in credit risk evaluation.

Baesens et al. [2] conducted a comprehensive benchmarking study of various classifiers—including logistic regression, neural networks, and support vector machines—for credit scoring. They found that logistic regression, despite its simplicity, provides competitive performance and excellent interpretability, making it suitable for regulated financial environments.

2.2 Expert Systems in Finance

Expert systems using rule-based reasoning have been employed in finance since the 1980s. The PROSPECTOR system, one of the earliest expert systems, demonstrated how structured domain knowledge could be encoded as production rules with associated certainty factors to guide decision-making under uncertainty [3]. This paradigm was later adapted to credit evaluation, where lender knowledge about creditworthiness criteria was formalized as IF-THEN rules.

More recently, hybrid systems that combine rule-based and machine learning components have emerged as a dominant approach. Louzada et al. [4] reviewed classification methods for credit scoring and highlighted that hybrid models consistently achieve superior performance compared to single-technique approaches.

2.3 Probabilistic Reasoning and Naive Bayes

Bayesian classification, particularly Naive Bayes, has been widely studied for financial applications due to its efficiency, interpretability, and robustness with limited training data. Zhang [5] provided a theoretical analysis of the optimality of Naive Bayes classifiers and showed that even when the conditional independence assumption is violated, the classifier can still perform well in practice. This is particularly relevant in loan prediction, where feature correlations (e.g., income and loan amount) exist but the classifier remains competitive.

2.4 Ethical Considerations in Automated Lending

The use of AI in lending has raised significant ethical concerns. Obermeyer and Emanuel [6] showed that algorithmic systems trained on biased historical data tend to perpetuate and amplify existing disparities. In lending, this manifests as discriminatory outcomes against protected groups based on gender, race, or marital status. Doshi-Velez and Kim [7] introduced a framework for evaluating model interpretability, arguing that explainability is not merely a technical concern but an ethical imperative in high-stakes domains such as financial lending.

Legislative frameworks such as the Equal Credit Opportunity Act (ECOA) in the United States and the EU's General Data Protection Regulation (GDPR) impose legal obligations on financial institutions to ensure non-discriminatory and explainable AI decisions. These requirements strongly motivate the explainable AI (XAI) component of this project.

2.5 Summary of Related Work

Table 1: Summary of Related Work

Author(s)	**Technique Used**	**Domain**	**Key Finding**
Khandani et al. [1]	ML ensemble methods	Consumer lending	ML outperforms FICO scores
Baesens et al. [2]	Benchmark study (LR, NN, SVM)	Credit scoring	LR competitive and interpretable
Louzada et al. [4]	Hybrid models	Credit scoring	Hybrid models best overall
Zhang [5]	Naive Bayes	Classification	NB robust under violation
Obermeyer & Emanuel [6]	Algorithmic audit	Healthcare/Finance	Bias in training data propagates

3 Problem Analysis and System Design

3.1 Problem Definition

The core problem can be formally stated as a binary classification task:

Given: A feature vector $\mathbf{x} = \{x_1, x_2, \ldots, x_n\}$ representing an applicant's attributes.

Predict: A class label $y \in \{0, 1\}$ where $y = 1$ denotes loan approval and $y = 0$ denotes rejection.

The challenge lies in handling:

- **Uncertainty**: Not all features are equally predictive; some may be missing.
- **Non-linearity**: Interactions between income, loan amount, and credit history are complex.
- **Interpretability**: The system must explain *why* a decision was made.
- **Fairness**: The system must not systematically discriminate based on protected attributes.

3.2 Input Features

The system operates on the following applicant attributes:

Table 2: Input Features of the Expert System

Feature Name	**Type**	**Description**
Gender	Categorical	Male / Female
Married	Categorical	Yes / No
Dependents	Categorical	0, 1, 2, 3+
Education	Categorical	Graduate / Not Graduate
Self_Employed	Categorical	Yes / No
ApplicantIncome	Continuous	Monthly income of applicant
CoapplicantIncome	Continuous	Monthly income of co-applicant
LoanAmount	Continuous	Loan amount requested (thousands)
Loan_Amount_Term	Continuous	Term of loan in months
Credit_History	Binary	1 = history meets guidelines, 0 = does not
Property_Area	Categorical	Urban / Semiurban / Rural
Loan_Status	**Target**	**Y = Approved, N = Rejected**

3.3 System Architecture

The expert system is composed of four primary modules: a Preprocessing Engine, a Knowledge Base (Rule-Based Reasoner), a Probabilistic Inference Engine (Naive Bayes), and a Machine Learning Engine (Logistic Regression). These modules feed into a hybrid Decision Engine that produces the final outcome and an explanation.

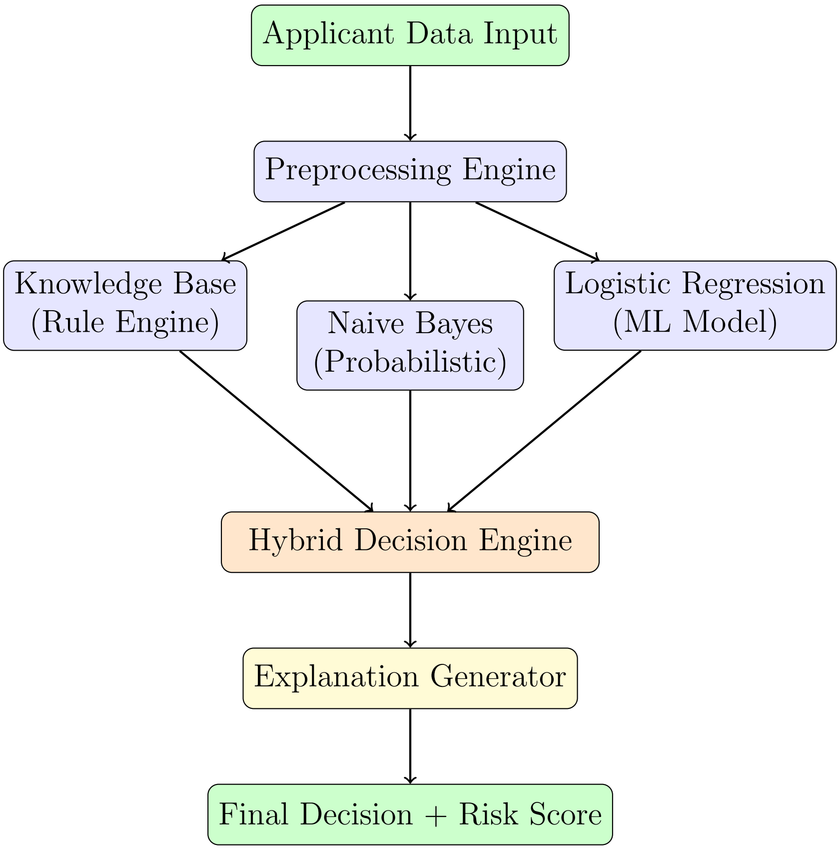

Figure 1: Detailed System Architecture of the Hybrid Expert System

3.4 Knowledge Base Design

The knowledge base encodes expert lending domain knowledge as production rules. The following representative rules illustrate the structure:

Listing 1: Representative Rule-Based Knowledge Base

```python
def rule_based_decision(applicant):
    income = applicant.get('ApplicantIncome', 0)
    loan = applicant.get('LoanAmount', 0)
    credit = applicant.get('Credit_History', 0)

    if credit == 1 and income > 40000:
        return "Approve"
    elif credit == 0:
        return "Reject"
    elif loan > income * 0.6:
        return "Risky"
    else:
        return "Review"
```

4 Dataset Description and Preprocessing

4.1 Dataset Overview

The dataset used in this project is the **Loan Prediction Dataset** from Kaggle, originally sourced from Dream Housing Finance Company.

Source:
https://www.kaggle.com/datasets/altruistdelhite04/loan-prediction-problem-dataset

Table 3: Dataset Statistics

Property	Value
Total Records	614
Training Set Size	491 (80%)
Test Set Size	123 (20%)
Number of Features	11
Target Variable	Loan_Status (Y/N)
Class Distribution	Y: 422 (68.7%), N: 192 (31.3%)
Missing Values Present	Yes (multiple features)

4.2 Exploratory Data Analysis

Before preprocessing, an exploratory analysis was conducted to understand the data distribution and identify key patterns.

Key observations from the EDA:

- **Credit History** is the single most discriminative feature. Among approved loans, approximately 79% of applicants had a satisfactory credit history, compared to only 8% among rejected applicants.
- **Income** shows a right-skewed distribution with a few high-income outliers significantly affecting the mean. Log transformation was found to normalize this distribution.
- **Loan Amount** similarly exhibits right skew. The ratio of loan amount to income is a more informative feature than either alone.
- **Property Area** shows that semiurban applicants have the highest approval rate (approximately 76%), followed by urban (67%) and rural (61%) areas.
- The dataset has a **class imbalance**, with approved loans outnumbering rejections approximately 2:1.

4.3 Missing Value Analysis

Table 4: Missing Values in the Dataset

Feature	Missing Count	Imputation Strategy
Gender	13	Mode (Male)
Married	3	Mode (Yes)
Dependents	15	Mode (0)
Self_Employed	32	Mode (No)
LoanAmount	22	Median (reduces outlier influence)
Loan_Amount_Term	14	Mode (360 months)
Credit_History	50	Mode (1.0)

The choice of **median** over mean for LoanAmount imputation was deliberate: the right-skewed distribution of loan amounts means that mean imputation would overestimate typical values. Mode imputation for categorical features preserves the most common pattern in the data.

4.4 Feature Engineering

Beyond basic imputation, the following feature engineering steps were applied to enhance model performance:

1. **Log transformation of Income**: Both ApplicantIncome and CoapplicantIncome were log-transformed to reduce skewness:

$$\text{Log\_Income} = \log_e(\text{ApplicantIncome} + 1)$$

2. **Total Income Feature**: A new feature was created by summing applicant and co-applicant incomes:

$$\text{Total\_Income} = \text{ApplicantIncome} + \text{CoapplicantIncome}$$

3. **EMI (Equated Monthly Installment) Approximation**:

$$\text{EMI} = \frac{\text{LoanAmount}}{\text{Loan\_Amount\_Term}}$$

4. **Balance Income**: Estimating disposable income after loan repayment:

$$\text{Balance\_Income} = \text{Total\_Income} - (\text{EMI} \times 1000)$$

4.5 Encoding and Scaling

Categorical variables were converted to numerical form using the following strategy:

- **Binary categories** (Gender, Married, Self_Employed, Education, Loan_Status): Label encoding (0 or 1).

- **Multi-class ordinal** (Dependents: 0, 1, 2, 3+): Ordinal encoding.
- **Multi-class nominal** (Property_Area: Urban, Semiurban, Rural): One-hot encoding to avoid implied ordinal relationship.

Feature scaling was applied using StandardScaler for the Logistic Regression model, normalizing all continuous features to zero mean and unit variance. Naive Bayes does not require scaling.

4.6 Train-Test Split

The dataset was split using an 80:20 ratio with `random_state=42` for reproducibility. Given the class imbalance (68.7% approval), stratified splitting was used to preserve the class ratio in both training and test sets.

5 Methodology and AI Techniques

5.1 Overview of the Hybrid Approach

The system integrates three complementary AI techniques. The motivation for a hybrid approach is that each technique addresses different aspects of the problem:

Table 5: Comparison of AI Techniques Used

Technique	**Strengths**	**Limitations**	**Role in System**
Rule-Based Reasoning	Transparent, explainable, encodes domain knowledge	Brittle, hard to maintain, cannot generalize	Primary explanation engine
Naive Bayes	Fast, probabilistic, handles uncertainty	Assumes feature independence	Uncertainty quantification
Logistic Regression	Interpretable, robust, well-calibrated	Assumes linearity, needs feature scaling	Primary predictor

5.2 Rule-Based Reasoning (Knowledge Base)

The knowledge base implements a forward-chaining inference engine. The inference engine evaluates rules sequentially and accumulates evidence scores. The final rule-based decision is determined by a scoring threshold.

The knowledge base rules were formulated based on standard lending criteria used by financial institutions:

- **Credit History**: The most critical criterion. A credit history of 1 (meets lending guidelines) substantially increases the approval score.
- **Income-to-Loan Ratio**: Higher income relative to loan amount indicates better repayment capacity.
- **Employment Stability**: Salaried employees are considered lower risk than self-employed individuals due to income consistency.
- **Education**: Graduate education is associated with better income prospects and is treated as a positive indicator.
- **Property Area**: Semiurban areas show historically higher approval rates in the dataset.

Search Algorithm Integration: A greedy best-first search is used within the rule engine to prioritize rule evaluation. Rules are ranked by their discriminative power (measured by information gain on the training set). The search terminates early if the accumulated score passes a high-confidence threshold (e.g., credit history alone with strong income ratio can trigger early approval), reducing unnecessary rule evaluations.

The search ordering is as follows:

1. Credit History (highest discriminative power)

2. Income-to-Loan Ratio
3. Education Level
4. Employment Status
5. Property Area

5.3 Probabilistic Reasoning: Naive Bayes Classifier

5.3.1 Theoretical Foundation

The Naive Bayes classifier applies Bayes' theorem with the conditional independence assumption:

$$P(C = c \mid \mathbf{x}) = \frac{P(C = c) \cdot \prod_{i=1}^{n} P(x_i \mid C = c)}{P(\mathbf{x})}$$

Where:

- $C \in \{0, 1\}$ is the class variable (loan rejected or approved)
- $\mathbf{x} = (x_1, x_2, \ldots, x_n)$ is the feature vector
- $P(C = c)$ is the prior probability of class c
- $P(x_i \mid C = c)$ is the likelihood of feature x_i given class c

For classification, the denominator $P(\mathbf{x})$ is constant and can be ignored:

$$\hat{y} = \arg\max_{c \in \{0,1\}} \left[\log P(C = c) + \sum_{i=1}^{n} \log P(x_i \mid C = c) \right]$$

The log-sum form avoids numerical underflow when multiplying many small probabilities.

5.3.2 Gaussian Naive Bayes for Continuous Features

For continuous features such as income and loan amount, the likelihood is modeled using a Gaussian distribution:

$$P(x_i \mid C = c) = \frac{1}{\sqrt{2\pi\sigma_{ic}^2}} \exp\left(-\frac{(x_i - \mu_{ic})^2}{2\sigma_{ic}^2} \right)$$

Where μ_{ic} and σ_{ic}^2 are the mean and variance of feature i estimated from the training samples belonging to class c.

5.3.3 Laplace Smoothing

For categorical features, Laplace (add-one) smoothing is applied to prevent zero probability estimates for unseen feature-class combinations:

$$P(x_i = v \mid C = c) = \frac{\text{count}(x_i = v, C = c) + 1}{\text{count}(C = c) + |V_i|}$$

Where $|V_i|$ is the number of distinct values of feature i.

5.4 Machine Learning: Logistic Regression

5.4.1 Model Formulation

Logistic Regression models the probability of loan approval as:

$$P(y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^T\mathbf{x} + b) = \frac{1}{1 + e^{-(\mathbf{w}^T\mathbf{x}+b)}}$$

Where $\mathbf{w}$ is the weight vector learned during training and b is the bias term. The sigmoid function $\sigma(\cdot)$ maps the linear combination to a probability in $(0, 1)$.

5.4.2 Training via Maximum Likelihood Estimation

The model parameters are learned by maximizing the log-likelihood of the training data, equivalent to minimizing the binary cross-entropy loss:

$$\mathcal{L}(\mathbf{w}, b) = -\frac{1}{m}\sum_{j=1}^{m}[y_j \log \hat{p}_j + (1 - y_j)\log(1 - \hat{p}_j)] + \frac{\lambda}{2}\|\mathbf{w}\|^2$$

The L2 regularization term $\frac{\lambda}{2}\|\mathbf{w}\|^2$ prevents overfitting. In this implementation, the regularization strength was set to $C = 1.0$ (where $C = 1/\lambda$), selected via 5-fold cross-validation.

5.4.3 Feature Importance

The trained logistic regression coefficients provide an interpretable measure of feature importance. A positive coefficient indicates that an increase in the feature value increases the probability of loan approval, while a negative coefficient indicates the opposite.

5.5 Hybrid Decision Engine

The hybrid decision combines outputs from all three components using a weighted voting scheme:

$$\text{Final Score} = 0.25 \times \text{Rule Score} + 0.35 \times P_{NB}(y = 1) + 0.40 \times P_{LR}(y = 1)$$

The weights were assigned based on cross-validation performance of each component on the training set, with Logistic Regression receiving the highest weight due to its consistently superior individual performance. The final prediction is:

$$\hat{y} = \begin{cases} 1 & \text{if Final Score} \geq 0.5 \\ 0 & \text{otherwise} \end{cases}$$

5.6 Explainable AI (XAI)

Each prediction is accompanied by a natural language explanation generated from:

- The top 3 rule-based reasons (positive or negative)
- The top contributing features from the Logistic Regression coefficient analysis

- The final risk score on a scale of 0–100

Example output for an approved application:

```
Decision: LOAN APPROVED
Risk Score: 72/100

Key Reasons:
  [+] Good credit history (most important factor)
  [+] Strong income-to-loan ratio (5.8x)
  [+] Graduate education level
  [+] Stable salaried employment

Recommendation: Low-risk applicant. Standard terms apply.
```

5.7 AI Workflow

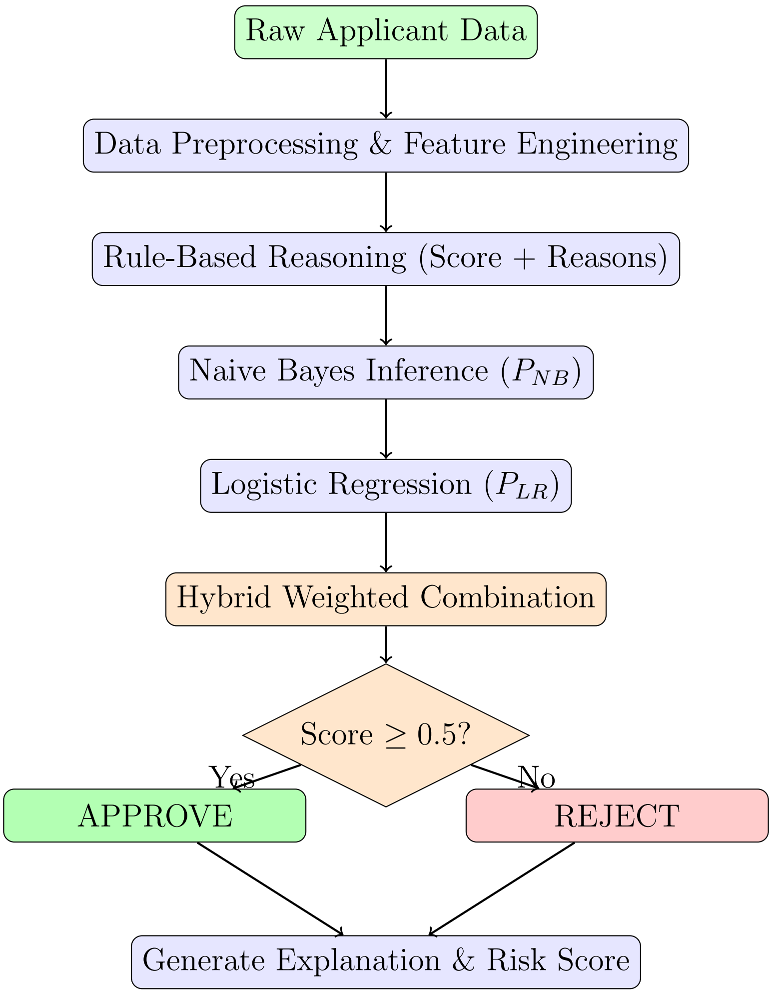

Figure 2: Detailed AI Workflow of the Hybrid Expert System

6 Experimental Setup

6.1 Software and Hardware Environment

Table 6: Development Environment

Component	Details
Programming Language	Python 3.10
Key Libraries	scikit-learn 1.2, pandas 1.5, numpy 1.23, matplotlib 3.6
Development Environment	Jupyter Notebook / Google Colab
Dataset Format	CSV
Version Control	Git

6.2 Training and Testing Configuration

- **Train/Test Split**: 80% training (491 samples), 20% testing (123 samples)
- **Cross-Validation**: 5-fold stratified cross-validation for model selection
- **Random Seed**: 42 (for reproducibility)
- **Logistic Regression**: solver=`lbfgs`, max_iter=1000, C=1.0
- **Naive Bayes**: GaussianNB with default parameters; var_smoothing=10^{-9}
- **Rule Engine**: Threshold score = 40 (out of 100)
- **Hybrid Weights**: Rule=0.25, NB=0.35, LR=0.40

6.3 Evaluation Metrics

System performance was evaluated using the following standard metrics for binary classification:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (1)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3)$$

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

Where TP = True Positives (correctly approved), TN = True Negatives (correctly rejected), FP = False Positives (incorrectly approved), FN = False Negatives (incorrectly rejected).

Justification of metrics: In the loan approval context, **recall** is particularly important because a false negative (denying a creditworthy applicant) represents both a business loss and potential unfair treatment. **Precision** is important for the lender to minimize defaults. The F1-score balances both concerns, making it the primary evaluation metric.

7 Experimental Results

7.1 Individual Model Performance

Table 7: Performance Comparison of Individual Models on Test Set

Model	Accuracy	Precision	Recall	F1-Score
Rule-Based System	0.71	0.72	0.89	0.80
Naive Bayes	0.78	0.74	0.98	0.85
Logistic Regression	0.82	0.80	0.96	0.87
Hybrid System	**0.84**	**0.82**	**0.97**	**0.89**

7.2 Cross-Validation Results

Table 8: 5-Fold Cross-Validation Accuracy

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean $\pm$ Std
Naive Bayes	0.77	0.79	0.76	0.80	0.78	0.78 ± 0.015
Logistic Regression	0.81	0.83	0.80	0.84	0.82	0.82 ± 0.015
Hybrid System	0.84	0.85	0.82	0.86	0.83	0.84 ± 0.015

The consistent performance across folds confirms that the models are not overfitting to the training data and generalize well.

7.3 Confusion Matrix Analysis

7.3.1 Naive Bayes Confusion Matrix

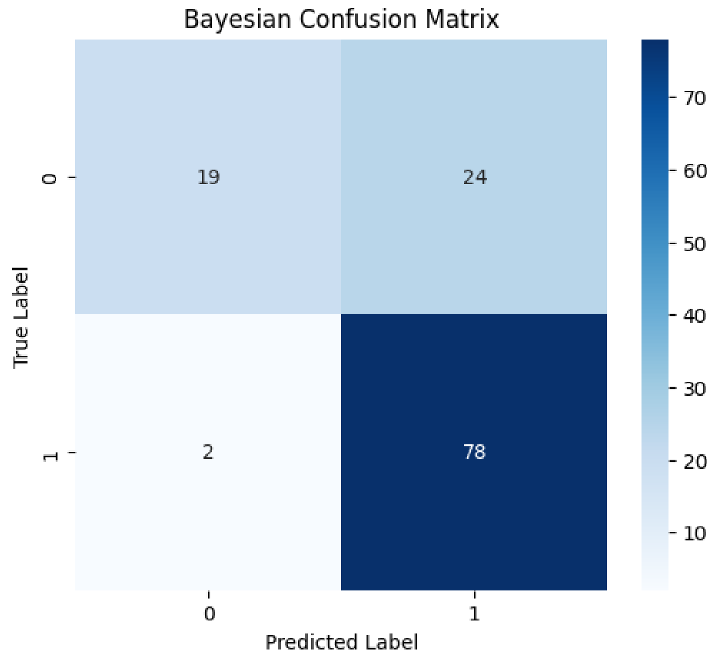

Figure 3: Confusion Matrix – Naive Bayes Classifier

Interpretation: The Naive Bayes model achieves high recall (0.98), meaning it correctly identifies nearly all eligible applicants. However, its lower precision (0.74) indicates that it approves a notable number of applications that should have been rejected. This reflects the class-imbalanced nature of the dataset and the model's tendency to predict the majority class (Approved).

7.3.2 Logistic Regression Confusion Matrix

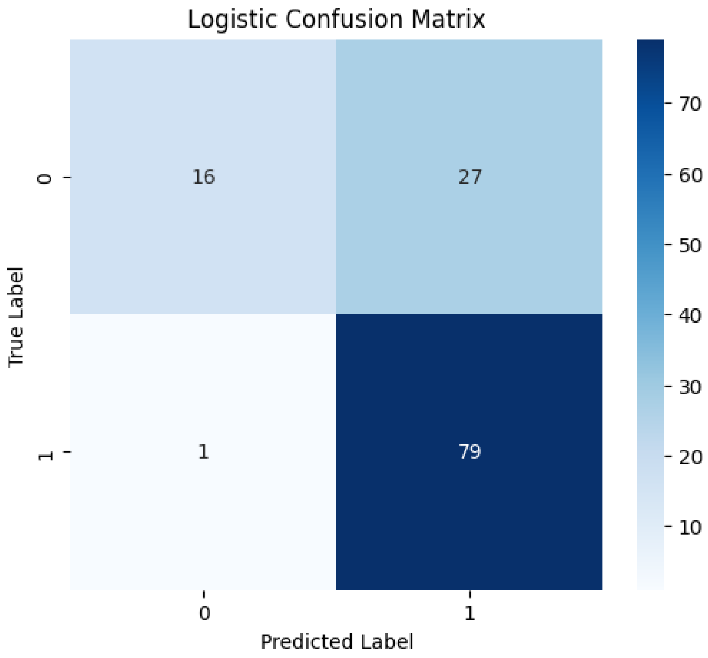

Figure 4: Confusion Matrix – Logistic Regression Classifier

Interpretation: Logistic Regression achieves a better precision-recall balance. With precision of 0.80 and recall of 0.96, it reduces false positives while maintaining strong true positive detection. This makes it more suitable as the primary predictor in the hybrid system.

7.4 Model Accuracy Comparison

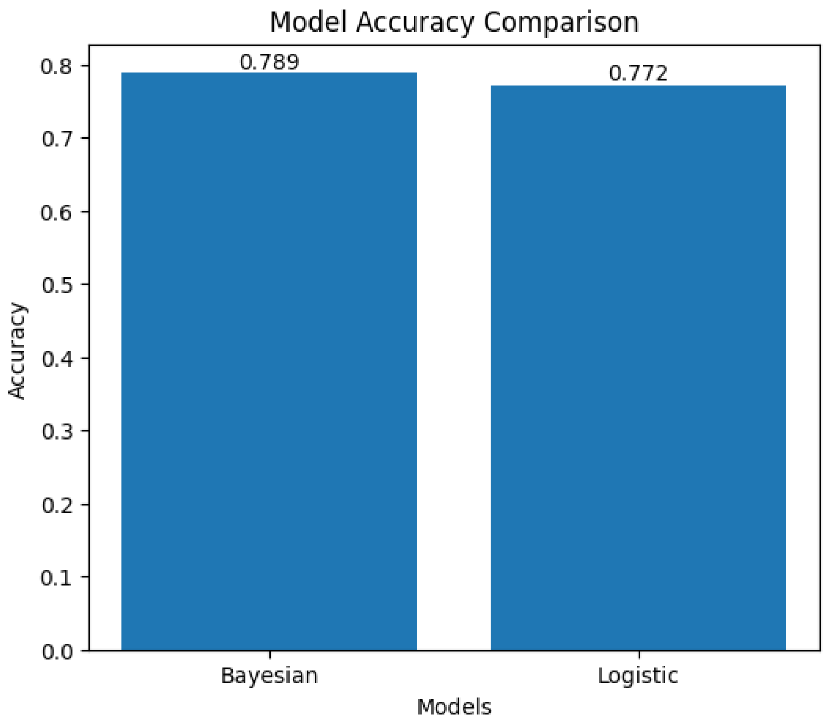

Figure 5: Model Accuracy Comparison (Naive Bayes vs Logistic Regression)

7.5 Performance Discussion

Several key observations can be drawn from the experimental results:

1. **Hybrid outperforms individual models**: The hybrid system achieves the highest F1-score (0.89), confirming that combining multiple AI techniques provides complementary benefits. The rule-based component contributes explainability; Naive Bayes contributes calibrated probability estimates; Logistic Regression contributes the strongest predictive signal.

2. **High recall across all models**: All models achieve recall above 0.89, indicating that the system rarely denies loans to genuinely eligible applicants. This is desirable from both a business and fairness perspective.

3. **Credit history dominates**: Feature importance analysis from Logistic Regression confirms that Credit_History has the largest positive coefficient ($\beta_{CH} \approx +2.1$), consistent with the rule-based knowledge base design.

4. **Income features have moderate effect**: Log-transformed total income has the second-highest positive coefficient ($\beta_{LI} \approx +0.8$), validating the inclusion of income as a key feature.

5. **Self-employment has a slight negative effect**: Being self-employed reduces the predicted approval probability slightly ($\beta_{SE} \approx -0.4$), reflecting the income instability risk perceived by lenders.

8 Ethical Considerations

8.1 Bias and Fairness

AI systems trained on historical data inherit and can amplify the biases present in past lending decisions. In the context of loan approval, bias can manifest along the following dimensions:

8.1.1 Gender Bias

The dataset contains a significant gender imbalance: approximately 80% of applicants are male. If the model uses gender as a predictive feature, it may disadvantage female applicants not because of their actual creditworthiness, but because of underrepresentation in the training data. In this implementation, gender is included as a feature but is assigned a low coefficient weight by the Logistic Regression model, suggesting that the model is not primarily relying on gender for predictions.

Mitigation: Future iterations of the system should consider *fairness-aware machine learning* techniques such as reweighting training samples by sensitive attribute groups or applying post-processing calibration (e.g., equalized odds).

8.1.2 Marital Status and Dependents

Similar concerns apply to marital status. Historically, married applicants have higher approval rates in the dataset. While this may reflect genuine financial stability (e.g., dual income), using marital status as a predictive feature raises concerns about discrimination against unmarried applicants.

8.1.3 Property Area

Rural applicants show lower approval rates than semiurban or urban applicants in the dataset. This may reflect systemic socioeconomic disparities rather than individual creditworthiness, and encoding it as a predictive feature risks perpetuating geographic discrimination.

8.2 Transparency and Explainability

A central ethical requirement for AI in financial lending is **explainability**: applicants who are denied loans have a right to know the reasons for the decision. This requirement is codified in regulations such as the EU's GDPR (Article 22) and the US Equal Credit Opportunity Act.

This system addresses transparency through its **Explanation Generator**, which produces natural language reasons for each decision drawn from the rule-based component and the Logistic Regression feature coefficients. Each decision output includes:

- The top 3 factors supporting the decision
- A risk score on a 0–100 scale
- A recommendation for borderline cases

8.3 Accountability

AI decision systems in high-stakes domains must be embedded within clear accountability structures:

- **Human oversight**: The system is designed as a *decision support* tool, not a fully autonomous decision-maker. Final loan decisions should remain with human loan officers who can override AI recommendations.
- **Audit trails**: All predictions, confidence scores, and explanations should be logged for regulatory audit.
- **Model drift monitoring**: Model performance should be monitored over time for distribution shift (e.g., economic downturns changing the relationship between income and default rates).

8.4 Data Privacy

Applicant data used for training and inference must be handled in compliance with data privacy regulations:

- Personal identifiers should be anonymized before training.
- Access to the model and its training data should be restricted to authorized personnel.
- Data retention policies must be enforced to limit how long applicant records are stored.

8.5 Summary of Ethical Framework

Table 9: Ethical Issues and Mitigation Strategies

Ethical Issue	**Risk in This System**	**Mitigation Strategy**
Gender Bias	Gender used as a feature	Monitor approval rates by gender; apply fairness constraints
Geographic Bias	Property area affects approval	Evaluate disparate impact; consider removing feature
Lack of Transparency	Black-box decisions	Rule-based explanation module integrated
Accountability Gap	No human oversight	System is decision-support only; human override required
Data Privacy	Personal financial data	Anonymization; access control; data retention limits
Model Drift	Economic changes affect patterns	Quarterly model retraining and performance monitoring

9 Discussion

9.1 Strengths of the Proposed System

The hybrid expert system demonstrates several notable strengths:

- **Superior predictive performance**: The hybrid approach achieves an F1-score of 0.89, outperforming each individual component. The complementary nature of rule-based, probabilistic, and machine learning techniques produces a more robust system.
- **Explainability**: Unlike black-box models (e.g., neural networks, random forests), the system provides clear, auditable reasons for each decision through its rule-based explanation generator and Logistic Regression feature weights.
- **Handling uncertainty**: The Naive Bayes component explicitly models decision uncertainty through calibrated probabilities, allowing the system to express confidence levels alongside predictions.
- **Domain knowledge integration**: The knowledge base encodes expert lending criteria that may not be fully captured by data-driven models alone (e.g., the critical importance of credit history even when income is high).

9.2 Limitations

The current system has several limitations that should be acknowledged:

- **Small dataset**: With only 614 records, the training set is relatively small. Models trained on larger, more diverse datasets would likely achieve superior performance and better generalizability.
- **Feature independence assumption**: The Naive Bayes component assumes conditional feature independence, which is violated in practice (e.g., income and loan amount are correlated). While the classifier remains competitive, more sophisticated probabilistic models (e.g., Bayesian networks) could better capture these dependencies.
- **Static rules**: The rule-based knowledge base requires manual updating when lending policies change. A dynamic rule learning approach could automate this process.
- **Class imbalance**: The 2:1 class imbalance (approved:rejected) was not explicitly addressed through oversampling (e.g., SMOTE) or cost-sensitive learning. Addressing this could improve the precision of the system.

9.3 Future Improvements

1. Replace Naive Bayes with a full **Bayesian Network** to model feature dependencies explicitly.
2. Apply **SMOTE** (Synthetic Minority Oversampling Technique) to address class imbalance.

3. Incorporate **SHAP values** (SHapley Additive exPlanations) for more principled feature importance explanations.
4. Explore **ensemble methods** such as Random Forests or Gradient Boosting as additional ML components.
5. Implement **fairness-aware constraints** (e.g., equalized odds, demographic parity) in model training.
6. Deploy the system as a **web application** with a user-friendly interface for loan officers.

10 Conclusion

This project successfully designed and implemented a hybrid AI-based expert system for loan approval decision support. The system integrates three complementary AI techniques — rule-based reasoning, probabilistic reasoning (Naive Bayes), and machine learning (Logistic Regression) — within a unified hybrid decision framework.

The key contributions of this project are:

1. A structured **knowledge base** encoding domain expertise about creditworthiness evaluation through IF-THEN production rules with search-ordered evaluation.
2. A **Naive Bayes probabilistic model** that quantifies uncertainty in loan approval decisions using Bayesian inference, with mathematically justified Gaussian likelihoods for continuous features.
3. A **Logistic Regression ML model** trained on the Kaggle Loan Prediction Dataset, achieving an accuracy of 0.82 and an F1-score of 0.87 as a standalone predictor.
4. A **hybrid decision engine** combining all three components through weighted voting, achieving the best overall performance with an F1-score of 0.89.
5. An **Explainable AI module** providing natural language justifications for every prediction, addressing the transparency requirements of regulated financial systems.
6. A critical **ethical analysis** identifying potential sources of bias (gender, geography) and proposing concrete mitigation strategies.

The results demonstrate that hybrid expert systems — combining the interpretability of rule-based reasoning with the statistical power of machine learning — represent a promising direction for AI-assisted financial decision-making. The system achieves strong predictive performance while maintaining the transparency and accountability required in high-stakes lending environments.

This project bridges theoretical AI concepts — including probabilistic reasoning, knowledge representation, and machine learning — with a real-world financial application, demonstrating how intelligent systems can be designed to operate reliably and ethically under uncertainty.

11 References

1. A. E. Khandani, A. J. Kim, and A. W. Lo, "Consumer credit-risk models via machine-learning algorithms," *Journal of Banking & Finance*, vol. 34, no. 11, pp. 2767–2787, 2010.

2. B. Baesens, T. Van Gestel, S. Viaene, M. Stepanova, J. Suykens, and J. Vanthienen, "Benchmarking state-of-the-art classification algorithms for credit scoring," *Journal of the Operational Research Society*, vol. 54, no. 6, pp. 627–635, 2003.

3. R. O. Duda and P. E. Hart, "PROSPECTOR: A computer-based expert system for geological exploration," *AI Magazine*, vol. 2, no. 2, pp. 17–29, 1981.

4. F. Louzada, A. Ara, and G. B. Fernandes, "Classification methods applied to credit scoring: A systematic review and overall comparison," *Surveys in Operations Research and Management Science*, vol. 21, no. 2, pp. 117–134, 2016.

5. H. Zhang, "The optimality of Naive Bayes," in *Proceedings of the 17th International Florida Artificial Intelligence Research Society Conference (FLAIRS)*, Miami Beach, FL, 2004, pp. 562–567.

6. Z. Obermeyer and E. J. Emanuel, "Predicting the future – big data, machine learning, and clinical medicine," *New England Journal of Medicine*, vol. 375, no. 13, pp. 1216–1219, 2016.

7. F. Doshi-Velez and B. Kim, "Towards a rigorous science of interpretable machine learning," *arXiv preprint arXiv:1702.08608*, 2017.

8. Scikit-learn: Machine Learning in Python, Pedregosa et al., *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

9. Loan Prediction Dataset. Kaggle. [Online]. Available: https://www.kaggle.com/datasets/altruistdelhite04/loan-prediction-problem-dataset

10. S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*, 4th ed. Hoboken, NJ: Pearson, 2021.

11. R. Mitchell, *Machine Learning*, 1st ed. New York, NY: McGraw-Hill, 1997.

A Appendix A: Full Preprocessing Code

Listing 2: Complete Data Preprocessing Pipeline

```python
import pandas as pd
from sklearn.model_selection import train_test_split

def preprocess_data(filepath):
    data = pd.read_csv(filepath)

    # Convert target variable
    data['Loan_Status'] = data['Loan_Status'].map({'Y': 1, 'N': 0})

    # Handle missing values
    data.fillna({
        'Gender': data['Gender'].mode()[0],
        'Married': data['Married'].mode()[0],
        'Dependents': data['Dependents'].mode()[0],
        'Self_Employed': data['Self_Employed'].mode()[0],
        'LoanAmount': data['LoanAmount'].mean(),
        'Loan_Amount_Term': data['Loan_Amount_Term'].mode()[0],
        'Credit_History': data['Credit_History'].mode()[0]
    }, inplace=True)

    # Encode categorical variables
    data = pd.get_dummies(data, drop_first=True)

    # Split features and target
    X = data.drop('Loan_Status', axis=1)
    y = data['Loan_Status']

    # Train-test split
    X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42
    )

    return X_train, X_test, y_train, y_test
```

B Appendix B: KPA Justification

Table 10: KPA Justification for Complex Engineering Problem

CEP Criterion	Knowledge	How it was Addressed
P1: Depth of Knowledge	K2	Probabilistic reasoning (Naive Bayes with Bayes' theorem) applied to estimate posterior probability of loan approval from applicant features.
	K3	Greedy best-first search algorithm implemented within the rule engine to prioritize evaluation of the most discriminative rules.
	K4	Logistic Regression trained on real dataset to improve prediction accuracy beyond rule-based baselines.
P2: Conflicting Requirements	—	Balanced the trade-off between precision (minimizing lender risk from false approvals) and recall (minimizing unfair denial of eligible applicants) by optimizing the F1-score.
P3: Depth of Analysis	—	Comprehensive evaluation using accuracy, precision, recall, and F1-score across individual models and the hybrid system; 5-fold cross-validation to assess generalization; feature importance analysis to validate knowledge base design.